Step 2a - H5 - Microwave access simulation

Market scan

Market size

The addressable pool is much wider than microwave OEMs: all industrial thermal-equipment manufacturers who could add microwave/RF modules to dryers, kilns, ovens and process lines. Adjacent markets are large — industrial dryers ~$5-8B/yr, industrial drying ovens ~$3B/yr — versus only ~$1.4-1.6B/yr currently captured by dedicated microwave equipment, which quantifies the headroom for microwave integration. The simulation/software slice an 'access' product could initially capture is small (~$20-100M/yr) but the mechanism is catalytic: lowering the barrier to entry expands the microwave market itself. Macro tailwind: process-heat electrification/decarbonization programs (US DOE Industrial Heat Shot, EU projects) explicitly name microwave systems.

Region	Size ($B)	Region	Size ($B)
Europe	~$1.5-3B adjacent thermal-equipment market (dryers+ovens subset); strong OEM density in DE/IT/FR	Asia	~$3-4B adjacent market (~40%+ of dryer demand; largest OEM pool, mostly mid-tier)
America	~$1.5-2.5B adjacent market	Africa	~$0.3-0.6B (incl. Middle East; mostly importers/integrators)

Target customers

Thousands of industrial equipment OEMs and system integrators (drying, thermal processing, food/pharma/chemical machinery) have no in-house RF/microwave competence. The motivated subset — those facing electrification pressure or customer pull for faster/volumetric heating — is realistically ~300-1,000 companies globally. They are reachable via general process-equipment channels and OEM-component suppliers, but each sale requires substantial education: the CalNEXT market study identifies 'knowledge and awareness' alongside cost and process-integration as the primary adoption barriers. A tool alone will not convert this segment; tool + application engineering service will.

Customer size	Approx. number	Market share (in %)
Large	~100-300 large equipment OEMs	~45-55% of adjacent equipment revenue
Medium	~1,000-3,000 mid-size machine builders	~30-40%
Small	~5,000-15,000 small builders/integrators	~10-20%

Players offering a solution today

No one sells 'microwave integration made accessible' as a product. What exists: (a) generalist EM/CFD suites at expert price/skill level (Ansys, CST, COMSOL), (b) cloud CAE platforms (SimScale, Ansys cloud) that remove the compute barrier but not the expertise barrier, (c) microwave component/generator vendors (Muegge, Sairem, Richardson, e2v) who effectively act as the integration path today via application labs and OEM supply, and (d) a fragmented layer of MW consultants and university labs. So the competitive field is indirect: ProSeed would compete with 'hire/borrow expertise' rather than with another software vendor.

Player	Type of solution	Market share (in %)
Large (Ansys/Synopsys, Dassault SIMULIA, COMSOL, Altair, Siemens, cloud CAE like SimScale)	generic multiphysics software + cloud HPC; powerful but require RF/EM specialists the target customers do not have	~70-80% of EM-sim spend; ~0% dedicated to non-expert MW integration
Medium (QWED QuickWave; MW component/generator vendors Muegge, SAIREM, Richardson, e2v with application labs; microwave design consultancies)	specialist simulation tools + de-facto 'integration service' via component supply and co-engineering	~10-20% niche
Small (independent MW/EM consultants, university groups via AMPERE/IMPI networks, open-source openEMS/MEEP)	one-off feasibility studies, contract design, free expert-level codes	~5-10%

Regulation

Regulation	Jurisdiction	What it means for us
IEC/EN 60519-6	International/EU	Safety standard for industrial microwave heating installations — an OEM integrating microwave modules must meet leakage/interlock requirements; a tool with built-in compliance checks directly de-risks their first product
CISPR 11 / EN 55011	International/EU	ISM emission limits; non-expert OEMs risk building non-compliant machines — guided templates with compliant reference designs are a selling point
ITU Radio Regulations ISM bands	Global, region-dependent	Restricts operating frequencies (915 MHz mainly Region 2, 2.45 GHz worldwide); an 'access' tool should hide band-selection complexity behind target-market presets
Machinery Directive 2006/42/EC + EMC Directive 2014/30/EU	EU	CE marking obligations for machines incorporating microwave subsystems; documentation support eases OEM compliance
Directive 2013/35/EU	EU	Worker EMF exposure limits shape enclosure/applicator requirements a novice OEM must respect

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	
Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis